

Cloud Computing Option 2: Serverless Image Processing Application on AWS Lambda

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Abstract

This report presents the design, implementation, deployment, and performance evaluation of a serverless image processing application developed using Amazon Web Services (AWS). The application allows users to upload images through a REST API and perform image processing operations using AWS Lambda functions. The deployment uses API Gateway, Lambda, Amazon S3, and CloudWatch services.

The main objective of the project is to evaluate the scalability and performance of serverless computing under different workload conditions. Experimental tests were conducted using varying numbers of concurrent users and requests. Metrics including response time, throughput, invocation count, execution duration, concurrency level, and error rate were collected and analyzed.

The obtained results demonstrate the ability of AWS Lambda to automatically scale according to workload intensity while maintaining acceptable performance levels. A cost analysis is also presented and compared with alternative deployment solutions.

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1 Introduction

1.1 Project Objectives

The objective of this project is to:

- Develop a serverless image processing application.
- Deploy the application using AWS Lambda.
- Evaluate performance under different workload conditions.
- Analyze scalability characteristics of serverless computing.
- Estimate operational costs and compare alternative deployment strategies.

1.2 Project Scope

The developed application supports:

- Image upload through REST APIs.
- Storage of images in Amazon S3.
- Image resizing operations.
- Monitoring through CloudWatch.
- Experimental performance evaluation.

2 System Design

2.1 Overall Architecture

Figure ?? presents the overall architecture of the proposed solution.

The workflow is summarized as follows:

1. User uploads an image.
2. API Gateway receives the request.
3. Lambda function processes the request.
4. Image is stored in Amazon S3.
5. Processed image is returned to the user.

2.2 AWS Components

2.2.1 API Gateway

API Gateway exposes REST endpoints used to upload and process images.

2.2.2 AWS Lambda

Lambda functions execute image processing operations without requiring dedicated servers.

2.2.3 Amazon S3

S3 stores both input and processed images.

2.2.4 CloudWatch

CloudWatch is used to collect monitoring and performance metrics during experimental testing.

3 Implementation

3.1 Upload Service

The upload endpoint receives an image encoded in Base64 format and stores it in Amazon S3.

Endpoint

/upload-base64

Example Request

```
{
  "filename": "test.jpg",
  "folder": "small",
  "content_type": "image/jpeg",
  "image_base64": "...
}
```

3.2 Processing Service

The processing endpoint performs image resizing operations.

Endpoint

/process

Example Request

```
{
  "bucket": "bucket-name",
  "key": "image.jpg",
  "operation": "resize",
  "width": 256,
  "height": 256
}
```

3.3 Deployment

The application was deployed within the AWS environment using:

- API Gateway
- AWS Lambda
- Amazon S3
- CloudWatch

Deployment configuration parameters are summarized in Table 1.

Table 1: Deployment Configuration

Parameter	Value
Lambda Runtime	TBD
Memory Allocation	TBD
Timeout	TBD
Region	TBD

4 Experimental Methodology

4.1 Evaluation Objectives

The evaluation focuses on the following metrics:

- Response Time
- Throughput
- Scalability
- Concurrent Executions
- Error Rate

4.2 Workload Scenarios

Four workload levels were considered.

Table 2: Workload Scenarios

Scenario	Concurrent Users
Small	10
Medium	50
Heavy	100
Stress	Until saturation

4.3 Testing Tools

The following tools were used:

- Apache JMeter
- AWS CloudWatch
- Postman

4.4 Metrics Collected

- Average Response Time
- Throughput
- Concurrent Executions

- Lambda Duration
- Invocation Count
- Error Rate

5 Experimental Results

5.1 Response Time Analysis

Figure 1 shows the relationship between workload intensity and response time.

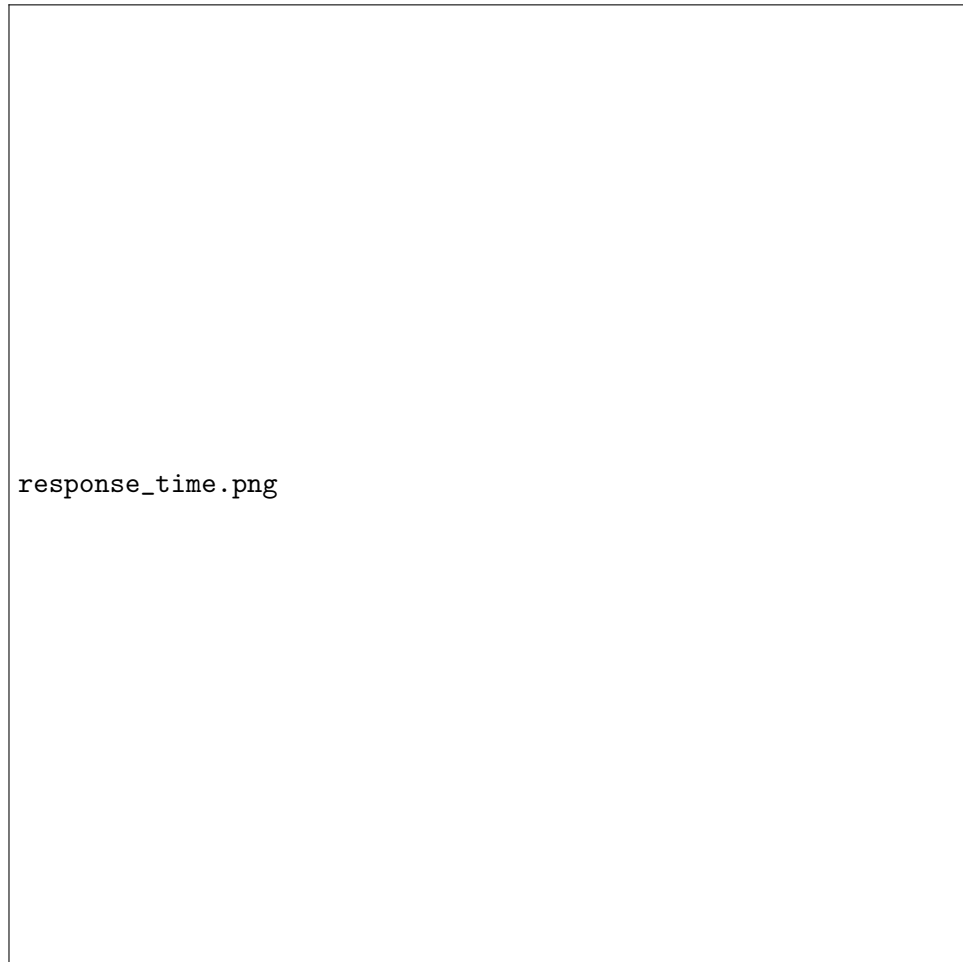


Figure 1: Response Time vs Concurrent Users

Discussion of observed trends goes here.

5.2 Throughput Analysis



Figure 2: Throughput vs Concurrent Users

Discussion of throughput behavior goes here.

5.3 Concurrency Analysis

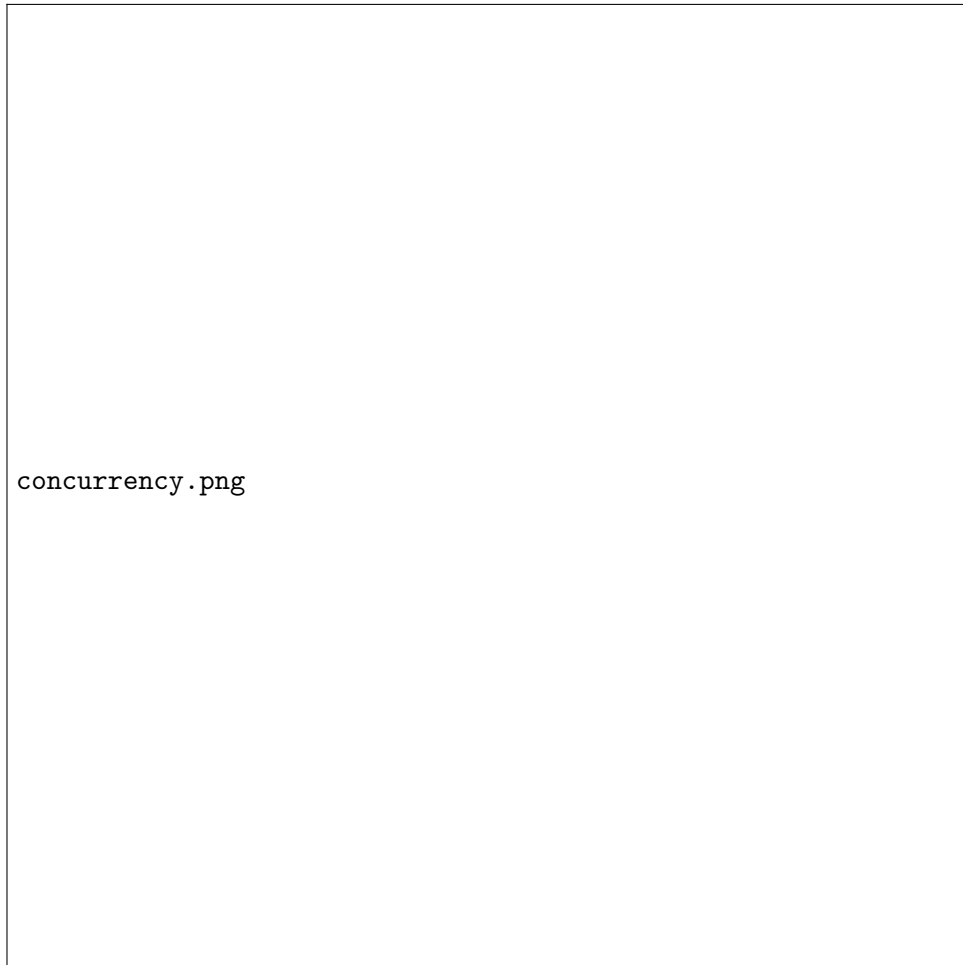


Figure 3: Concurrent Lambda Executions

CloudWatch observations and discussion go here.

5.4 Scalability Evaluation

This section discusses:

- Automatic scaling behavior.
- Maximum observed concurrency.
- Performance bottlenecks.
- Resource utilization.

5.5 Error Analysis

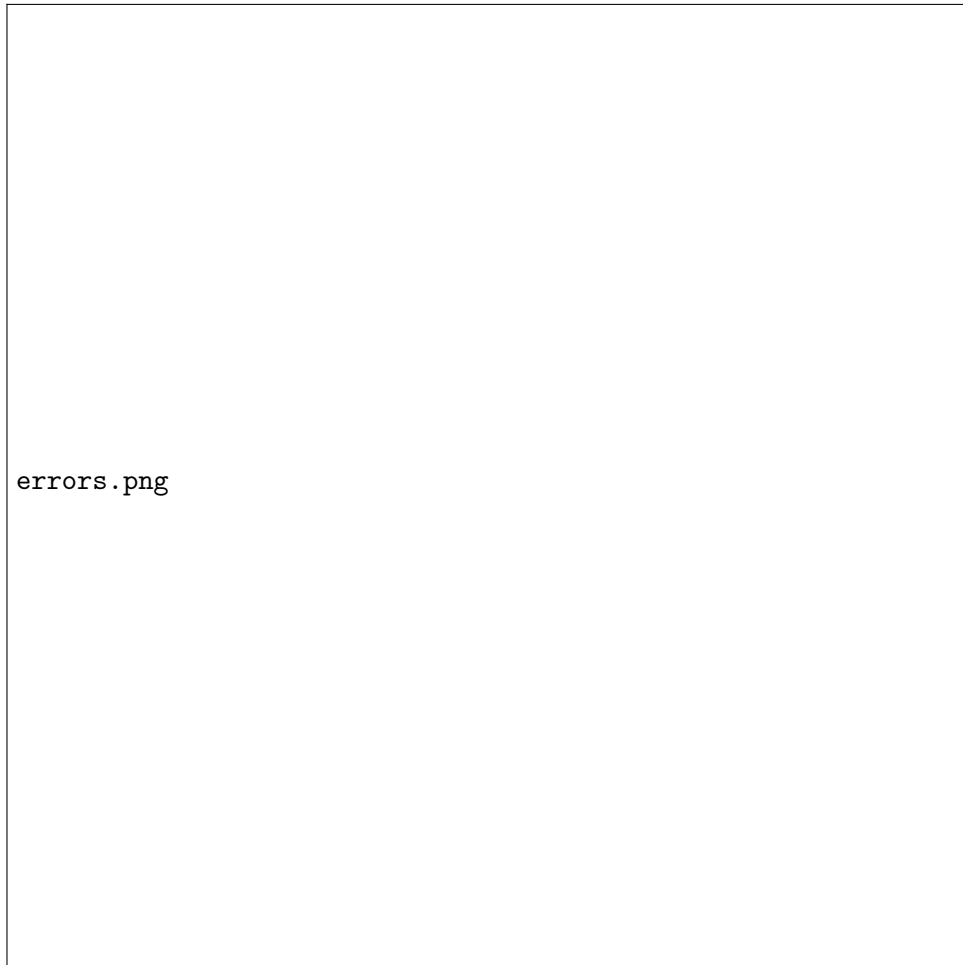


Figure 4: Error Rate vs Concurrent Users

Discussion of system reliability under load.

6 Cost Evaluation

6.1 AWS Lambda Cost Estimation

A six-month operational cost estimate is provided.

Table 3: Estimated Operational Cost

Component	Monthly Cost	6-Month Cost
Lambda	TBD	TBD
API Gateway	TBD	TBD
S3 Storage	TBD	TBD
CloudWatch	TBD	TBD
Total	TBD	TBD

6.2 Alternative Deployment Comparison

The proposed serverless solution is compared with an EC2-based deployment.

Table 4: Deployment Comparison

Metric	Lambda	EC2
Cost	TBD	TBD
Scalability	TBD	TBD
Management Effort	TBD	TBD
Availability	TBD	TBD

6.3 Discussion

Advantages and disadvantages of both approaches are discussed in this section.

7 Discussion

The experimental results indicate:

- Whether the system scales effectively.
- The workload levels at which degradation begins.
- The effectiveness of serverless deployment.
- Observed limitations and bottlenecks.

8 Conclusion

This project demonstrated the deployment of a serverless image processing application using AWS Lambda. Experimental evaluation showed how the system behaves under increasing workload intensity and highlighted the scalability characteristics of serverless computing. The results collected through CloudWatch and JMeter provide quantitative evidence of system performance. Finally, cost analysis demonstrated the economic implications of adopting a serverless architecture compared to traditional deployment solutions.

References

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- [2] Apache JMeter Documentation. <https://jmeter.apache.org>
- [3] Cloud Computing Course Material.